COMPLETE SQL LEARNING ROADMAP

An 8-week, 56-day hands-on plan

Learn one topic per day • Practise in MySQL Workbench • Document progress on GitHub

Goal	Build SQL skills from fundamentals through joins, analytical queries, database programming, and portfolio projects.
Daily deliverables	Notes, practice.sql, questions.sql, screenshots, and a Git commit.
Database	MySQL, using MySQL Workbench for hands-on practice.

How to use this roadmap

Work through the days in order. Try each query yourself before checking a solution. If a topic takes more than one day to understand, take the extra time. Consistency and understanding matter more than speed.

GitHub safety: use fictional data only. Never publish passwords, tokens, .env files, or private company data.

Week 1 — SQL Fundamentals

Day	Topic	What you'll learn / practical task
01	SQL and databases	SQL basics, databases, tables, rows, columns, and MySQL Workbench
02	Data types	INT, VARCHAR, DATE, DECIMAL, BOOLEAN
03	Database and table creation	CREATE DATABASE, CREATE TABLE, USE, DROP
04	Inserting data	INSERT INTO and inserting multiple rows
05	Retrieving data	SELECT, SELECT *, selecting specific columns, DISTINCT
06	Filtering data	WHERE, comparison operators, AND, OR, NOT
07	More filtering	IN, BETWEEN, LIKE, IS NULL, IS NOT NULL

Week 2 — Querying and Summarising Data

Day	Topic	What you'll learn / practical task
08	Sorting and limiting	ORDER BY, ASC, DESC, LIMIT
09	Updating and deleting	UPDATE, DELETE, and safe use of WHERE
10	Aggregate functions	COUNT, SUM, AVG, MIN, MAX
11	Grouping data	GROUP BY and grouping by multiple columns
12	Filtering groups	HAVING and WHERE vs HAVING
13	Conditional logic	CASE, IF, IFNULL, COALESCE
14	SQL revision	Mixed queries and a mini assessment

Week 3 — Relationships and Joins

Day	Topic	What you'll learn / practical task
15	Keys and constraints	Primary keys, foreign keys, UNIQUE, NOT NULL, DEFAULT, CHECK
16	Relationships	One-to-one, one-to-many, and many-to-many
17	INNER JOIN	Matching records between tables
18	LEFT JOIN	All left-table records and matching records
19	Other joins	RIGHT JOIN, cross joins, and self joins
20	Multi-table joins	Joining three or more tables
21	Join practice	Solve business-style questions using joins

Week 4 — Intermediate SQL

Day	Topic	What you'll learn / practical task
22	String functions	CONCAT, LENGTH, UPPER, LOWER, TRIM, SUBSTRING
23	Numeric functions	ROUND, ABS, CEIL, FLOOR, MOD
24	Date functions	CURDATE, DATEDIFF, DATE_ADD, YEAR, MONTH
25	Subqueries	Queries inside other queries
26	Subqueries with operators	EXISTS, NOT EXISTS, ANY, ALL
27	Set operations	UNION and UNION ALL
28	Revision	Combined filtering, grouping, functions, and joins

Week 5 — Advanced SQL

Day	Topic	What you'll learn / practical task
29	Altering tables	ALTER TABLE, add/modify/drop columns, rename columns
30	Views	CREATE VIEW, CREATE OR REPLACE VIEW, DROP VIEW
31	Common Table Expressions (CTEs)	WITH, multiple CTEs, and recursive CTE basics
32	Window functions	ROW_NUMBER, RANK, DENSE_RANK
33	Window calculations	OVER, PARTITION BY, LAG, LEAD
34	Indexes	CREATE INDEX, DROP INDEX, and index basics
35	Query execution	EXPLAIN and understanding query execution plans

Week 6 — Database Programming and Reliability

Day	Topic	What you'll learn / practical task
36	Transactions	START TRANSACTION, COMMIT, ROLLBACK
37	Savepoints	SAVEPOINT, ROLLBACK TO SAVEPOINT, RELEASE SAVEPOINT
38	Stored procedures	CREATE PROCEDURE, parameters, CALL
39	Stored functions	CREATE FUNCTION and return values
40	Control flow	IF, ELSEIF, CASE, and loops in stored programs
41	Triggers	CREATE TRIGGER, BEFORE, AFTER, INSERT, UPDATE, DELETE
42	Prepared statements	Parameters, PREPARE, EXECUTE, and SQL injection prevention concepts

Week 7 — Practical Projects and Interview Preparation

Day	Topic	What you'll learn / practical task
43	Database design	Design tables for a real-world application
44	Normalisation	Understand 1NF, 2NF, 3NF, and reduce duplication
45	Constraints and relationships	Build a relational database with valid relationships
46	E-commerce project	Customers, products, orders, and order items
47	Employee analytics project	Salaries, departments, rankings, and aggregations
48	Data analysis questions	Business questions using joins, CTEs, and window functions
49	Interview practice	Beginner, intermediate, and advanced SQL questions

Week 8 — Revision and Portfolio

Day	Topic	What you'll learn / practical task
50	Basic SQL revision	Solve 15 questions without notes
51	Filtering and functions	Solve 15 query problems
52	Joins and subqueries	Solve 15 relationship-based questions
53	Aggregation and window functions	Solve 15 analytical questions
54	Debugging SQL	Fix incorrect queries and explain the errors
55	Project completion	Finish one SQL project with a clean README
56	Mock interview	Attempt a timed SQL assessment and review mistakes

Daily GitHub Workflow

For each day, keep the work organised in a topic-specific folder:

```
Day-01-SQL-Basics/
■■■ notes.md
■■■ practice.sql
■■■ questions.sql
■■■ screenshots/
    ■■■ results.png
```

Daily checklist

■ Learn the concept and write short notes in your own words.

■ Execute the examples in MySQL Workbench.

■ Attempt extra questions without copying the solution first.

■ Save SQL files and a screenshot of the results.

■ Review errors, fix them, and record what you learned.

■ Commit and push the day's folder to GitHub.

Example Git commands

```
git status
git add Day-01-SQL-Basics
git commit -m "Complete Day 01 SQL basics"
git push
```

Important notes

• This roadmap targets MySQL; syntax may differ in PostgreSQL, SQL Server, or SQLite.

• Practise UPDATE, DELETE, and DROP only in your own learning database.

• Prepared statements are one protection against SQL injection; learn parameterised queries in your application code too.

• The roadmap is a study plan, not a guarantee of job readiness. Aim to explain your query choices clearly.